SILVER OAK UNIVERSITY

School of Technology, Design & Computer Application

Bachelor of Technology

Computer Science & Engineering

Course Name: Software Engineering

Course Code:2010043336

Semester:5th

Prerequisite:

Object Oriented Programming and Fundamentals of UML.

Objectives:

1. To study Software Development Life Cycle and various software process models including Agile methodologies.
2. To understand software requirements engineering and design software systems using Object Oriented and Function Oriented approaches.
3. To apply software project management techniques including cost estimation, scheduling and risk management.
4. To analyze and evaluate software testing strategies, techniques and test case design.
5. To study software quality standards, DevOps practices and modern software engineering tools.

Teaching Scheme:

Teaching Scheme				
L	T	P	Contact Hours	Credit
3	0	2	5	4

Contents:

Unit	Topics	Teaching Hours	Weightage %
1	**UNIT:1 INTRODUCTION TO SOFTWARE ENGINEERING AND AGILE DEVELOPMENT:**Software Engineering as a Layered Technology, Software Process Models — Waterfall, Prototyping, RAD, Spiral, Software Product vs Software Process, Agile Process Model, Scrum, Kanban, Extreme Programming (XP), Agile Requirements Engineering.	08	19
2	**UNIT:2 SOFTWARE REQUIREMENT MODELING AND SPECIFICATION:** Requirements Development Methodology, Software Requirements Specification (SRS), Eliciting Requirements, Requirements Modeling, Requirements Validation and Traceability, Managing Changing Requirements.	05	12

3	**UNIT:3 SOFTWARE ANALYSIS AND DESIGN:** Design Concepts and Design Principles, Architectural Design, Function Oriented Design, Object Oriented Design, User Interface Design, Web Application Design.	06	14
4	**UNIT:4 SOFTWARE PROJECT MANAGEMENT:** W5HH of Project Management, Software Metrics — Process Metrics, Product Metrics and Project Metrics, Software Cost and Effort Estimation, Project Scheduling and Tracking, Risk Analysis and Management.	05	12
5	**UNIT:5 SOFTWARE CODING AND TESTING:** Coding Standards and Guidelines, Code Review, Software Testing Life Cycle (STLC), Testing Strategies — Unit, Integration, Regression, System and Acceptance Testing, Testing Techniques — Black Box and White Box Testing, Test Case Design.	06	14
6	**UNIT:6 SOFTWARE QUALITY ASSURANCE, MAINTENANCE AND CONFIGURATION MANAGEMENT:** Software Quality Concepts, Software Quality Assurance (SQA), Software Configuration Management (SCM), Types of Software Maintenance, CMMI, ISO 9001.	05	12
7	**UNIT:7 DEVOPS AND SE TOOLS:** DevOps — Principles, Practices and Lifecycle, CI/CD Pipeline, Continuous Testing and Continuous Monitoring, Version Control — GitHub, UI/UX Design Tool — Figma, Code Quality Tool — SonarQube, Testing Tool — JUnit, Project Management Tool — JIRA.	07	17

Course Outcomes:

Sr. No.	CO Statement	Unit
CO-1	Describe software process models and Agile methodologies.	1
CO-2	Analyze Software Requirements and SRS document.	2
CO-3	Design software systems using Function Oriented and Object Oriented approaches.	3
CO-4	Analyze software cost estimation and risk management techniques.	4
CO-5	Evaluate software testing strategies and test cases.	5
CO-6	Summarize software quality standards, DevOps practices and SE tools.	6,7

Teaching & Learning Methodology:

The various methods or tools to teach the above subject:

1. Problem - based learning
2. Design Thinking
3. Cooperative-based learning
4. Competency-based learning
5. Case Study based Learning
6. Tool based Learning

List of Experiments: **Total Hours: 28**

Sr. No.	Practical Name
1	Develop a comprehensive case study on creation of a web or mobile application using Agile software engineering methodology.
2	Create Software Requirements Specification (SRS) document for any one domain — Healthcare, Education or E-commerce.
3	Develop wireframes and mockups for a web or mobile application using Figma.
4	Create Use Case Diagrams and Activity Diagrams using draw.io for any one domain.
5	Develop Class Diagrams and Sequence Diagrams using draw.io for any one domain.
6	Create Software Project Management Plan (SPMP) including effort estimation and scheduling using GanttProject or JIRA.
7	Illustrate project workflow using GitHub to showcase version control and collaboration practices.
8	Develop detailed test cases and scenarios for any one domain — Healthcare, Education or E-commerce.
9	Perform software quality review and inspection using code review techniques and prepare SQA report.
10	Bug tracking and issue management using JIRA to illustrate task management and progress tracking.

Books Recommended:

1. Roger S. Pressman, “Software engineering- A practitioner’s Approach”, McGraw -Hill International Editions
2. Ian Sommerville, “Software engineering”, Pearson educationAsia
3. Pankaj Jalote, “Software Engineering” – A Precise Approach Wiley
4. Ali Behhforoz & Frederick Hudson OXFORD, “Software Engineering Fundamentals”.
5. Rajib Mall, “Fundamentals of software Engineering”, Prentice Hall of India.
6. Deepak Gaikwad, Viral Thakkar, “DevOps Tools from Practitioner’s ViewPoint”, Wiley
7. Armando Fox and David Patterson ,”Engineering Software as a Service An Agile Software Approach”.
8. John M Nicolas, “Project Management for Business”, Engineering and Technology, Elsevier

List of Open Source Software/learning website:

1. https://devops.com/most-popular-open-source-devops-tools/
2. www.onesmartclick.com/engsineering/software-engineering.html
3. www.sei.cmu.edus

CO-PO-PSO Matrix:

Co. No.	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO-1	2	1			1				1	1		1		1
CO-2	3	2	1	2	1				2	1		1		2
CO-3	2	3	3	1	2				1	1		1	1	2
CO-4	2	3	2	3	1	1		2	2	1	3	1		2
CO-5	2	3	2	3	2			1	1	1	1	2	1	2
CO-6	1	1	2	1	3	1		1	2	2	1	3	2	1